

$GL_3(\mathbb{F}_2)$

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Problem 1. Show that $X^3 + 1 = (x + 1)(X^2 + X + 1)$, $(X + 1)^3 = X^3 + x^2 + x + 1$, $X^3 + X^2 + 1$, and $X^3 + X + 1$ are the possible characteristic polys for such a matrix check that all these polys except for $(x + 1)^3$ have distinct roots in any field containing $\mathbb{F}_2$ – recall you can check for double roots (in some bigger field) by checking whether the poly and its formal derivative have a common factor (we will reprove this this term). Recall also from last term the cool fact that if F inside G is a subfield, then two square matrices over F are conjugate over F iff they are conjugate over G . Conclude that if we fix one of the polys other than $(x + 1)^3$, all the matrices with this char poly are conjugate. So we get three conjugacy classes corresponding to those three polys

Proof. Any cubic polynomial in $\mathbb{F}_2$ would be of the form, $x^3 + ax^2 + bx + c$, where $a, b, c \in \mathbb{F}_2$. Thus the only polynomials would be:

1. $x^3 + 1 = (x + 1)(X^2 + X + 1)$
2. $X^3 + x^2 + x + 1 = (X + 1)^3$
3. $X^3 + X^2 + 1$
4. $X^3 + X + 1$

To check for double roots we take the second derivatives and check for common linear factors.

1. $\frac{d}{dx}(x^3 + 1) = 3x^2$, no common linear factors. Thus no double roots for polynomial 1. Thus over a larger field containing $\mathbb{F}_2$ matrices with this characteristic polynomial are diagonalizable, because there are distinct roots, thus conjugate, because they have the same eigenvalues. By last semester, we know if matrices are conjugate in a larger field they are conjugate in the smaller feild, in our case $\mathbb{F}_2$.
2. Checking for $(x + 1)^3$ is redundant because it already has 1 root with multiplicity 3.
3. $\frac{d}{dx}(X^3 + X^2 + 1) = 3x^2 + 2x = x(3x + 1)$, no common linear factors. Thus no double roots for polynomial 3. Thus over a larger field containing $\mathbb{F}_2$ matrices with this characteristic polynomial are diagonalizable, because there are distinct roots, thus conjugate, because they have the same eigenvalues. By last semester, we know if matrices are conjugate in a larger field they are conjugate in the smaller field, in our case $\mathbb{F}_2$.
4. $\frac{d}{dx}(X^3 + X + 1) = 3x^2 + 1$, no common linear factors. Thus no double roots for polynomial 4. Thus over a larger field containing $\mathbb{F}_2$ matrices with this characteristic polynomial are diagonalizable, because there are distinct roots, thus conjugate, because they have the same eigenvalues. By last semester, we know if matrices are conjugate in a larger field they are conjugate in the smaller field, in our case $\mathbb{F}_2$.

Thus if we fix poly 1, 3, or 4, all matrices with this characteristic poly are conjugate as stated above, so we get 3 conjugacy classes corresponding to this polynomial. $\square$

Problem 2. Explain why there are exactly three conjugacy classes of matrices with $(x+1)^3$ as their poly – consider the jordan forms (and use our fact that conjugate over a bigger field is the same as conjugate over the given field). Recall if a matrix has all the roots of its characteristic poly, then it has a Jordan form over the given field.

Proof. There are 3 conjugacy classes corresponding to the matrices with $(x+1)^3$ as their characteristic polynomial, because they correspond to the possible jordan forms, any matrix with this characteristic polynomial has a jordan form which its conjugate to. The possible jordan forms are I , $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, and

$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Since every matrix with this characteristic poly, has a jordan form thus is conjugate to one of these jordan forms. Thus each of the jordan forms create 3 more conjugacy classes. $\square$

Problem 2b. Representative elements for each conjugacy class and orders of size of conjugacy class.

Proof. 1. $X^3 + x^2 + x + 1 = (X + 1)^3$

(a) I

(b) $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Suppose $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \in \mathbb{Z}(A)$. Then $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} A = A \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, thus $\begin{bmatrix} a+d & b+e & c+f \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & a+b & c \\ d & d+e & f \\ g & g+h & i \end{bmatrix}$. Thus $a = e, d = f = g = 0$. $\begin{bmatrix} a & b & c \\ 0 & a & 0 \\ 0 & h & i \end{bmatrix} \in Z(A)$.

But since the determinant must be 1, $a^2 i = 1 \implies a = i = 1$. Thus the centralizer of A consists of matrices, $\begin{bmatrix} 1 & b & c \\ 0 & 1 & 0 \\ 0 & h & 1 \end{bmatrix}$ where $b, c, h \in \mathbb{F}_2$. Thus there are 8 elements in the centralizer. By orbit stabilizer, $[A] = \frac{168}{8} = 21$.

(c) $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Suppose $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \in \mathbb{Z}(B)$. Then $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} B = B \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Thus $\begin{bmatrix} a+d & b+e & c+f \\ d+g & h+e & f+i \\ g & h & i \end{bmatrix} = \begin{bmatrix} a & a+b & b+c \\ d & d+e & f+e \\ g & g+h & h+i \end{bmatrix}$. Thus $d = g = h = 0, a = e = 1, f = b$.

Thus the matrices in $Z(B)$ have the form, $\begin{bmatrix} a & b & c \\ 0 & a & b \\ 0 & 0 & a \end{bmatrix}$, however since the determinant is

1, $a^3 = 1 \implies a = 1$. Thus the matrices look like, $\begin{bmatrix} 1 & b & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$. Thus $|Z(B)| = 4$ and

$[B] = 168/4 = 42$.

2. $x^3 + 1 = (x+1)(X^2 + X + 1)$. A representative element is $C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$, Suppose $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \in \mathbb{Z}(C)$. Then $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} C = C \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Thus $\begin{bmatrix} a & b & c \\ g & h & i \\ d+g & h+e & f+i \end{bmatrix} = \begin{bmatrix} a & c & b+c \\ d & f & f+e \\ g & i & h+i \end{bmatrix}$. Thus $b = c = d = g = 0$, $f = h$, $i = f + e$. However since commuting matrices preserve the eigenspaces, $a = 1$. Thus the matrices have the form, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & e & f \\ 0 & f & e+f \end{bmatrix}$ where the determinant is 1, thus $e^2 + ef - f^2$, this is nonzero unless $e = f = 0$. Thus there are 3 choices and $|Z(C)| = 3 \implies [C] = \frac{168}{3} = 56$

3. $x^3 + x + 1$. $D = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Suppose $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \in \mathbb{Z}(D)$. Then $\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} D = D \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Thus $\begin{bmatrix} g & h & i \\ a+g & b+h & c+i \\ a+d & b+e & c+f \end{bmatrix} = \begin{bmatrix} b+c & c & a+b \\ f+e & f & d+e \\ h+i & i & g+h \end{bmatrix}$. Thus matrices have the form, $\begin{bmatrix} a & b & c \\ b & a & b+c \\ b+c & c & a+b \end{bmatrix}$ and the determinant must be 1, so $a(a(a+b) - c(b+c)) - b(b(a+b) - (b+c)^2) + c(bc - ab - ac) = a(a^2 + ab - bc - c^2) - b(ab + b^2 - b^2 - 2bc - c^2) + c(bc - ab - ac) = a(a^2 + ab - bc - c^2) - b(ab - 2bc - c^2) + c(bc - ab - ac) = a^3 + a^2b - abc - ac^2 - ab^2 + 2b^2c + bc^2 + bc^2 - abc - ac^2 = a^3 + a^2b = a^3 - ab^2 + a^2b - 2ac^2 + 2bc^2 + 2b^2c - 2abc$ which is nonzero unless $a = b = c = 0$. Thus there are 7 possibilities for the matrices. Thus $Z(D) = 7$ and $[D] = \frac{168}{7} = 24$

4. $X^3 + X^2 + 1$. Representative element is $D = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$. This must be the last conjugacy class thus $[E] = 168 - [A] - [B] - [C] - [D] = 24$.

□

Problem 3. Consider the action on $X = F_2^3 \setminus 0$, the natural permutation action. Show that the trace of an element g (for the action of G on C^X) is the number of eigenvectors. Subtract off the triv, and see you get Artin's χ_6 .

Proof. $\forall M \in GL_3(\mathbb{F}_2)$, suppose $x \in F_2^3 / \text{setminus } 0$ is an eigenvector of M , thus $\lambda \in \mathbb{F}_2 \ni Mx = \lambda x$. However λ can either be 0 or 1, but since M is invertible and $x \neq 0$, $Mx \neq 0$, thus $\lambda = 1$. Thus x is a fixed point of M . Similarly suppose x is a fixed vector of M , then $Mx = x \implies x$ is a 1 eigenvector of M .

For any permutation matrix, the trace gives the number of fixed points a matrix has. Thus $\forall g \in GL_3(\mathbb{F}_2)$, $\text{tr}(g)$ is the number of fixed fixed vectors of g , which is equal to the number of 1 eigenvectors, since a fixed point is a 1 eigenvector of g . Thus the character of the action of X , is the following:

	(1) [I]	(21) [A]	(42) [B]	(56) [C]	(24) [D]	(24) [E]
$\chi(X)$	7	3	1	1	0	0
$\chi_6 = \chi(X) - \chi_{triv}$	6	2	0	0	-1	-1

□

Problem 4. Consider the action of G on X . Show its "doubly transitive" ie acts transitively on pairs of distinct elements. Show that the stabilizer for the action on pairs is iso to F_2^2 (the vector space, as additive group). Now consider the action on X and the stablizer G^x . Show this has a surjective map to $GL_2(F_2)$ (by considering the quotient of F_2^3 by the subspace spanned by x). Show again the kernel is F_2^2 . Prove the stabilizer G^x is S_4 . When we have talked about induced reps. this will give us a way to come up with some further reps (taking reps of S_4 and looking at the "induced" reps).

Proof. Let $G := GL_3(F_2)$ $Y := \{(x, y) \in F_2^3 \times F_2^3 | x \neq y\}$. $\forall (x, y), (x', y') \in Y$, since $x \neq 0, y \neq 0, x' \neq 0, y' \neq 0$ x and y are linearly independent and x' and y' are linearly independent. Choose $z, z' \in X$, such that $span(x, y, z) = F_2^3$ and $span(x', y', z') = F_2^3$. Thus $\begin{bmatrix} x & y & z \end{bmatrix}$ and $\begin{bmatrix} x' & y' & z' \end{bmatrix}$ are in $GL_3(F_2)$. By closure of groups, $\exists M \in GL_3(F_2) \ni M = \begin{bmatrix} x' & y' & z' \end{bmatrix} \begin{bmatrix} x & y & z \end{bmatrix}^{-1}$, thus $M \begin{bmatrix} x & y & z \end{bmatrix} = \begin{bmatrix} x' & y' & z' \end{bmatrix}$ and $Mx = x'$ and $My = y'$. Thus $M(x, y) = (x', y')$. Thus M is doubly transitive.

Let $x = (e_1, e_2)$. $G^x = \{g \in GL_3(F_2) : g(e_1, e_2) = (e_1, e_2)\}$. Thus $\forall g \in G^x, g = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & c \end{bmatrix}$,

however since $g \in GL_3(\mathbb{F}_2)$, $c = 1$. Thus $g = \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$. There are 4 possible pairs for a and b .

Thus $|G^x| = 4$. Let $\phi : F_2^2 \rightarrow G^x$ where $\begin{bmatrix} a \\ b \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix}$. $\forall \begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} c \\ d \end{bmatrix} \in F_2^2, \phi(\begin{bmatrix} a \\ b \end{bmatrix})\phi(\begin{bmatrix} c \\ d \end{bmatrix}) =$

$\begin{bmatrix} 1 & 0 & a \\ 0 & 1 & b \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & c \\ 0 & 1 & d \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & a+c \\ 0 & 1 & b+d \\ 0 & 0 & 1 \end{bmatrix} = \phi(\begin{bmatrix} a \\ b \end{bmatrix} + \begin{bmatrix} c \\ d \end{bmatrix})$. Thus ϕ is a homomorphism

$\forall v = \begin{bmatrix} a \\ b \end{bmatrix} \in \ker \phi$, it means $a = 0$ and $b = 0$, since $\phi(v) = I$, so $a = [I]_{1,3} = 0$ and $b = [I]_{2,3} = 0$. Thus $\ker \phi = 0$ and ϕ is injective. Since $|F_2^2| = 4 = |G^x|$ and since ϕ is injective, the map is a isomorphism. Thus $G^x \cong F_2^2$.

$\forall x \in X, F_2^3 / span(x) \cong F_2^2$. $\forall g \in G^x, g = \begin{bmatrix} 1 & * & * \\ 0 & a & b \\ 0 & c & d \end{bmatrix}$, the block $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL_2(F_2)$. Thus there is a

map from $\phi : G^x \rightarrow GL_2(F_2)$, where $\begin{bmatrix} 1 & * & * \\ 0 & a & b \\ 0 & c & d \end{bmatrix} \rightarrow \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. $\forall A = \begin{bmatrix} 1 & * & * \\ 0 & a & b \\ 0 & c & d \end{bmatrix}, B = \begin{bmatrix} 1 & * & * \\ 0 & e & f \\ 0 & g & h \end{bmatrix} \in G^x$.

$\phi(AB) = \phi\left(\begin{bmatrix} 1 & * & * \\ 0 & bg+ae & af+bh \\ 0 & dg+ce & cf+dh \end{bmatrix}\right) = \begin{bmatrix} bg+ae & af+bh \\ dg+ce & cf+dh \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \phi(A)\phi(B)$. Thus

ϕ is a homomorphism. $\forall \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in GL_2(F_2), \exists g = \begin{bmatrix} 1 & * & * \\ 0 & a & b \\ 0 & c & d \end{bmatrix} \in GL_3(F_2) \ni \phi(g) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Thus

ϕ is surjective. Let $\rho : \mathbb{F}_2^2 \rightarrow \ker \phi$, where $\begin{bmatrix} a \\ b \end{bmatrix} \rightarrow \begin{bmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \rho\left(\begin{bmatrix} a+c \\ b+d \end{bmatrix}\right) = \begin{bmatrix} 1 & a+c & b+d \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} =$

$\begin{bmatrix} 1 & a & b \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & c & d \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \rho\left(\begin{bmatrix} a \\ b \end{bmatrix}\right)\rho\left(\begin{bmatrix} c \\ d \end{bmatrix}\right)$. Thus ρ is homomorphism. $|F_2^2| = |\ker \phi| = 4$. $\forall k \in \ker \rho$, $k = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, thus $\ker \rho = 0$, and ρ is injective. Thus ρ is isomorphism.

$GL_2(F_2)$ acts on $\{(e_1, e_2), (e_1, e_1 + e_2), (e_2, e_1 + e_2)\}$. Let k be the permutation which fixes all the basis, thus $ke_1 = e_1$ and $ke_2 = e_2$, thus $k = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Thus $k = I$, thus the action is faithful. Thus $GL_2(F_2) \leq S_3$ but $|GL_2(F_2)| = |S_3| = 6$, thus $GL_2(F_2) \cong S_3$.

Thus $G^x \cong F_2^2 \rtimes GL_2(F_2) \cong K_4 \rtimes S_3$. $S_3 \cong GL_2(F_2) \rightarrow GL_2(F_2)$, is so we send every automorphism of F_2 to itself. Let $y = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $z = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. Thus $G^x \cong \langle e_1, e_2, y, z | e_1^2 = e_2^2 = y^3 = z^2 = 1, e_1e_2 = e_2e_1, ye_1y^{-1} = e_2, ye_2y^{-1} = e_1e_2, ze_1z^{-1} = e_1, ze_2z^{-1} = e_2 \rangle \cong S_4$, by universal properties because S_4 also has these relations. $\square$

Problem 5. Show that G has 8 sylow 7 subgroups

Proof. $|G| = 168 = 2^3 * 3 * 7$.

The number of sylow 7 are divide 24 and are 1 mod 7. The only possibilities are 1 and 8 however if it was 1, the sylow 7 would be normal, however no disjoint union of conjugacy classes gives a set of size 7. Thus the sylow 7 cannot be normal and thus the number of sylow 7s are 8. $\square$